

Name: _____

General Instructions:

1. Do not turn this page until instructed.
2. Duration: 70 minutes. Total: 60 points.
3. Read carefully. Ask the professor if in doubt.
4. Write clearly. Justify every answer.
5. Do not detach pages. Simple calculator allowed.
6. Answers without development will not be credited.

Student's signature and ID

QUESTION	MAX SCORE	OBTAINED
1	20	
2	20	
3	20	
TOTAL	60	
GRADE		

Question 1.

(20 points)

Consider the set $A = \{x \in \mathbb{R} : |x^2 - 4| < 3(x + 2)\}$.

- a) Determine the solution set A in interval notation. (12 pts)
- b) Determine $\inf A$, $\sup A$, and decide whether A has a minimum and/or maximum. Justify each claim formally. (8 pts)

Question 2.

(20 points)

Context. To finance the *Plan Chile Sale Adelante*, the government levied a transitory tax on the mining sector, which absorbed **74% of the total fiscal cost** of the plan (Clapes UC, March 2026; Law 21.811, *Diario Oficial*, March 26, 2026). Simultaneously, the removal of the diesel subsidy — diesel rose **\$580 CLP per liter** (ENAP, *Informe de Precios Estimados*, March 25, 2026) — elevated operational costs for copper mining, where diesel represents approximately **40% of total expenses** (Cochilco, *Cost Observatory*, 2025). As a mine increases production, it exhausts its highest-grade ore and must process progressively lower-grade material, consuming exponentially more diesel per pound of copper produced.

Let $x \geq 0$ represent a mining operation's annual copper production in **millions of pounds** of fine copper. The revenue and cost functions are:

$$R(x) = 4.00x \quad C(x) = 0.05x^2 + 1.50x + 20$$

where both functions are in **millions of USD**. The revenue uses a spot price of **USD 4.00/lb** on the London Metal Exchange (Cochilco, 2025). The cost reflects: the quadratic fuel penalty from ore grade depletion ($0.05x^2$); baseline variable costs ($1.50x$); and fixed capital costs plus the operation's proportional share of the transitory tax (20).

- a) Determine all values of $x \geq 0$ for which the operation generates a **strictly positive net profit**, i.e., $R(x) > C(x)$. Express your answer in interval notation. (12 pts)
- b) Let $A = \{x \geq 0 : R(x) > C(x)\}$ be the set of viable annual production levels. Determine $\inf A$, $\sup A$, and decide whether A has a minimum and/or maximum. Justify each claim formally, and interpret $\inf A$ and $\sup A$ in one sentence each in terms of the mining operation. (8 pts)

Question 3.

(20 points)

Context. On March 26, 2026, the Chilean government removed the MEPCO fuel subsidy, raising the price of 93-octane gasoline by **\$392 CLP per liter** — the largest single-week fuel price adjustment in Chilean history (ENAP, *Informe de Precios Estimados*, March 25, 2026). Because Chile imports **97% of its fuel**, ENAP acts as a price-taker on international markets, making the supply curve perfectly horizontal (ENAP, official methodology). The weekly fuel market for 93-octane gasoline in Santiago is modeled as follows, where x is weekly consumption in **millions of liters** and $D(x)$, $S(x)$ are in **CLP per liter** (calibrated from elasticity $\varepsilon_d = -0.23$, Carrasco 2012, Universidad de Chile; ENAP pre-intervention price, March 19, 2026):

$$D(x) = 6143.6 - 99.9x \quad S(x) = 1148.8 \quad x \geq 0$$

Key concepts (read carefully before solving). **Shortage:** $D(x) > S(x)$ — consumers are willing to pay more than producers require; demand cannot be fully satisfied at the current price. **Surplus:** $D(x) < S(x)$ — supply exceeds what consumers demand. **Equilibrium:** the quantity x^* where $D(x^*) = S(x^*)$: supply and demand balance exactly. **Government intervenes** when the shortage $D(x) - S(x)$ exceeds a threshold $M > 0$ CLP/liter, signaling critical market stress.

- a)** Verify that the pre-intervention equilibrium occurs at $x^* = 50$ million liters/week at $P^* = \$1,148.8$ CLP/liter. Determine all $x \geq 0$ for which there is a **shortage** and all $x \geq 0$ for which there is a **surplus**. (5 pts)
- b)** The government must choose an intervention threshold $M > 0$ CLP/liter. Determine the **intervention zone** $I(M) = \{x \geq 0 : D(x) - S(x) > M\}$ and the **non-intervention shortage zone** $N(M) = \{x \geq 0 : D(x) > S(x) \text{ and } D(x) - S(x) \leq M\}$, both in terms of M . For what values of M is $N(M)$ **non-empty**? Find the value of M such that $N(M)$ has **length exactly 10** million liters/week. (8 pts)
- c)** Using the value of M found in part (b), determine $\inf N(M)$, $\sup N(M)$, and decide whether $N(M)$ has a minimum and/or maximum. Justify each claim formally, and interpret $\inf N(M)$ and $\sup N(M)$ in one sentence each in terms of the fuel market. (7 pts)